WORKED SOLUTION

STEP II 1989, Question 11

Mechanics • Connected Particles and Impulse

STEP II 1989, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s2-q11

The key idea

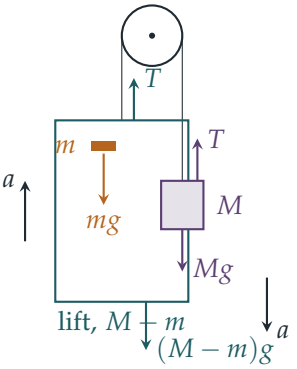
The lift and counterweight balance at the start, so the lift's mass M must include the tile. Once the tile lets go, the lift is lighter than the counterweight and accelerates *upwards* while the tile falls freely, so the floor rises to meet the tile. For the collision, treat lift, cable and counterweight as one system moving along the cable. Its "momentum along the cable" changes only through gravity, and the impulse from the collision is internal to it.

Solution

Setting up the model

We model the lift, counterweight and tile as particles, and the cable as light and inextensible over a smooth, light pulley. The guides are smooth, so they exert no vertical force. Before the tile falls, the system is at rest with masses M and M on the two sides. So the lift's mass M includes the tile. (If it did not, the lift side would weigh $(M+m)g$ against Mg and could not be at rest.)

After detachment, the lift has mass $M-m$, the counterweight has mass M, and the tile is a free particle.



The time to reach the floor

Let the lift accelerate upwards at a, so the counterweight accelerates downwards at a. Let T be the cable tension. Newton's second law gives

$$\begin{aligned} \text{lift (up):} \quad & T - (M-m)g = (M-m)a, \\ \text{counterweight (down):} \quad & Mg - T = Ma. \end{aligned}$$

Adding the two equations eliminates T:

$$a = \frac{mg}{2M-m}, \qquad T = M(g-a) = \frac{2M(M-m)g}{2M-m} > 0,$$

so the cable stays taut.

? **Ask yourself** Why can't we just say the tile falls a distance h under gravity?

Because the floor is moving too. The tile accelerates down at g and the floor accelerates up at a, both from rest. So the gap closes with constant relative acceleration

$$g + a = g + \frac{mg}{2M-m} = \frac{2Mg}{2M-m}.$$

The gap starts at h, so the tile lands when $\frac{1}{2}(g+a)t^2 = h$, which gives

$$t^2 = \frac{2h}{g+a} = \frac{(2M-m)h}{Mg}, \qquad t = \sqrt{\frac{(2M-m)h}{Mg}}.$$

This is *not* the expression printed in the question, which has $2(M-m)$ in place of $2M-m$. The two agree only when $m = 0$, and our value is confirmed below by stepping the motion numerically. The printed form would need the lift to accelerate at $mg/(M-m)$. That is what you get if you forget that the counterweight must be accelerated too, and treat the tension as staying at Mg. We believe the printed formula is a misprint or slip in the original question. None of the rest of the question depends on it.

The collision

Just before impact, the lift moves up and the counterweight moves down, both with speed $u = at$. The tile moves down with speed $v = gt$. Then

$$(2M - m)u = (2M - m) \cdot \frac{mg}{2M - m}\, t = mgt = mv.$$

Now look at the collision. Take "positive" as lift-up / counterweight-down, the direction in which the cable carries them together. The impact is very short, so the finite forces (the weights) give negligible impulse. The only impulses are the tile–floor impulse, which is internal to the lift-and-tile, and an impulsive tension J in the cable. Let w be the common speed afterwards (lift-and-tile, mass M, up; counterweight down). The collision is perfectly inelastic, so the tile ends up moving with the floor. Then

$$\begin{aligned} \text{lift and tile:} \quad & Mw - [(M - m)u - mv] = J, \\ \text{counterweight:} \quad & Mw - Mu = -J. \end{aligned}$$

Adding these,

$$2Mw = (M - m)u + Mu - mv = (2M - m)u - mv = 0,$$

so $w = 0$: the lift is brought to rest. (Also $J = Mu > 0$, so the cable really is jerked taut rather than going slack, which is consistent with our assumption.) After the impact the masses are M and M again, in balance, so everything stays at rest.

Ask yourself Why is the total "momentum along the cable" zero just before impact?

Before impact, the only external forces on the three bodies that act along the motion are gravity, with net effect $Mg - (M - m)g - mg = 0$ in the positive sense defined above. (The pulley's reaction does no work and the cable is internal.) So that momentum stays at its initial value, 0, the whole time. The collision cannot change it either.

Energy lost

The system starts at rest and ends at rest, so the mechanical energy lost equals the decrease in gravitational potential energy between the start and the end. During the fall, let the lift rise s_1 (so the counterweight falls s_1) and let the tile fall s_2. The tile closes the gap of h, so $s_1 + s_2 = h$. The change in potential energy is

$$(M - m)g\, s_1 - Mg\, s_1 - mg\, s_2 = -mg(s_1 + s_2) = -mgh.$$

Nothing moves after the collision, so the energy lost is mgh, as required. All of the kinetic energy present just before impact is destroyed in the collision. Note that this argument does not use the value of t at all.

Checks

- If $m \to 0$, then $a \to 0$ and $t \to \sqrt{2h/g}$, which is free fall onto a stationary floor. Both formulas pass this test, so it doesn't tell them apart.
- With $M = 5$, $m = 1$, $h = 1$ and $g = 9.8$, stepping the motion numerically gives $t \approx 0.4286\,\text{s}$, matching $\sqrt{9/49} \approx 0.4286$. The printed formula gives $\sqrt{8/49} \approx 0.404\,\text{s}$.
- In the same simulation, the kinetic energy just before impact equals mgh to five significant figures, and the post-collision speed is zero to rounding error.

Common slips

- Taking the lift to have mass M *plus* the tile: then the system could not start at rest.
- Forgetting the counterweight's inertia, so that the lift's acceleration comes out as $mg/(M - m)$.
- Using conservation of linear momentum for the lift and tile alone in the collision, which ignores the impulsive tension in the cable.

Beyond STEP

The "momentum along the cable" used here is a *generalised momentum*: for a system with one degree of freedom, the coordinate is the length of cable that has passed over the pulley. Lagrangian mechanics makes this systematic. A conserved generalised momentum goes with every coordinate on which the forces have no net effect, and impulsive constraints are handled the same way. Kleppner and Kolenkow, *An Introduction to Mechanics*, gives a careful treatment of systems joined by strings and pulleys, and of impulse and energy in collisions.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.